

Advanced Portfolio Theory: Mathematical Foundations and Modern Developments

A Rigorous Treatment for Mathematicians and Statisticians

Arthur Mota

University of Sao Paulo

June 2025 (revised October 2026)

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Course Objectives

Primary Goals

- Rigorous mathematical treatment of portfolio theory
- Bridge classical results with modern developments
- Emphasis on measure-theoretic foundations
- Applications of advanced optimization and statistical methods

Prerequisites

- Measure theory and functional analysis
- Stochastic processes and martingale theory
- Convex optimization
- Statistical inference and asymptotic theory

Historical Context and Modern Relevance

Timeline of Portfolio Theory:

- **1952:** Markowitz mean-variance theory
- **1964–66:** CAPM (Sharpe, Lintner, Mossin)
- **1969–71:** Merton's continuous-time theory
- **1979–81:** Harrison–Kreps, Harrison–Pliska fundamental theorems
- **1990s:** Coherent risk measures
- **2000s:** High-dimensional methods
- **2010s:** Machine learning integration

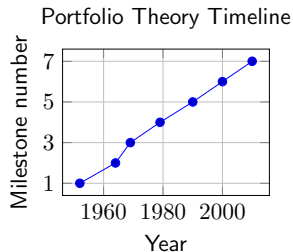


Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations**
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Probability Spaces and Filtrations

Definition (Filtered Probability Space)

A **filtered probability space** is a tuple $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ where:

- $(\Omega, \mathcal{F}, \mathbb{P})$ is a complete probability space
- $\{\mathcal{F}_t\}_{t \geq 0}$ is a filtration
- $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for $0 \leq s \leq t$

Usual Conditions

- 1 **Right-continuity:** $\mathcal{F}_t = \bigcap_{s > t} \mathcal{F}_s$
- 2 **Completeness:** $\mathcal{F}_0$ contains all $\mathbb{P}$ -null sets

Stochastic Processes: Definitions

Definition (Adapted Process)

A stochastic process $X = \{X_t\}_{t \geq 0}$ is **adapted** to $\{\mathcal{F}_t\}$ if X_t is $\mathcal{F}_t$ -measurable for all $t \geq 0$.

Definition (Predictable Process)

A process X is **predictable** if it is measurable with respect to the predictable σ -algebra on $\Omega \times [0, \infty)$.

Theorem (Doob-Meyer Decomposition)

Every càdlàg submartingale Y of class (DL) can be uniquely decomposed as:

$$Y_t = M_t + A_t$$

where M is a right-continuous martingale and A is a predictable increasing process with $A_0 = 0$.

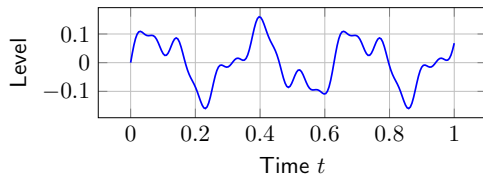
Brownian Motion

Definition (Standard Brownian Motion)

A process $W = \{W_t\}_{t \geq 0}$ is standard Brownian motion if:

- 1 $W_0 = 0$ almost surely
- 2 W has independent increments
- 3 $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$
- 4 W has continuous paths

Schematic illustration (not a simulated path)



Smooth stand-in curve, for illustration only: a true Brownian path is nowhere differentiable.

Theorem (Itô's Lemma)

Let X_t satisfy $dX_t = \mu_t dt + \sigma_t dW_t$ and $f \in C^{1,2}([0, T] \times \mathbb{R})$. Then:

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma_t \frac{\partial f}{\partial x} dW_t$$

Itô Isometry

For predictable H with $\mathbb{E} \left[\int_0^T H_s^2 ds \right] < \infty$:

$$\mathbb{E} \left[\left(\int_0^T H_s dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^T H_s^2 ds \right]$$

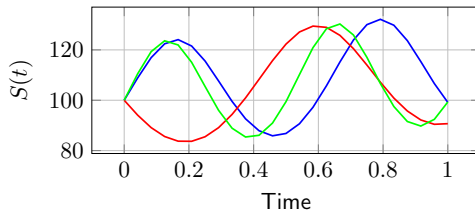
Geometric Brownian Motion

Definition (Geometric Brownian Motion)

The process $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$ satisfies:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Schematic paths from a common $S_0 = 100$ (not simulated)



Smooth stand-in curves, for illustration only: simulated GBM paths are nowhere differentiable.

Martingale Theory

Definition (Martingale)

A process $M = \{M_t\}_{t \geq 0}$ is a **martingale** if:

- 1 M is adapted and integrable
- 2 $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$ for all $0 \leq s \leq t$

Theorem (Optional Stopping Theorem)

If M is a right-continuous martingale and τ is a stopping time, with τ bounded or M uniformly integrable, then:

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$$

Martingale Representation

Theorem (Martingale Representation Theorem)

Let $\{\mathcal{F}_t\}$ be the augmented filtration generated by the Brownian motion W . Every square-integrable $\mathcal{F}_T$ -measurable random variable H can be represented as:

$$H = \mathbb{E}[H] + \int_0^T \phi_s dW_s$$

where ϕ is predictable with $\mathbb{E} \int_0^T \phi_s^2 ds < \infty$.

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory**
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

The Markowitz Problem

Portfolio Optimization Problem

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w}$$

subject to:

$$\boldsymbol{\mu}^T \mathbf{w} = \mu_p \quad (1)$$

$$\mathbf{1}^T \mathbf{w} = 1 \quad (2)$$

where:

- $\mathbf{w} \in \mathbb{R}^n$ is the portfolio weight vector
- $\Sigma \in \mathbb{R}^{n \times n}$ is the covariance matrix
- $\boldsymbol{\mu} \in \mathbb{R}^n$ is the expected return vector

Lagrangian Solution

The Lagrangian:

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma) = \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} - \lambda (\boldsymbol{\mu}^T \mathbf{w} - \mu_p) - \gamma (\mathbf{1}^T \mathbf{w} - 1)$$

First-order conditions yield:

$$\mathbf{w} = \lambda \Sigma^{-1} \boldsymbol{\mu} + \gamma \Sigma^{-1} \mathbf{1}$$

Let:

- $A = \mathbf{1}^T \Sigma^{-1} \mathbf{1}$
- $B = \mathbf{1}^T \Sigma^{-1} \boldsymbol{\mu}$
- $C = \boldsymbol{\mu}^T \Sigma^{-1} \boldsymbol{\mu}$
- $D = AC - B^2$

Efficient Frontier

Theorem (Efficient Frontier Equation)

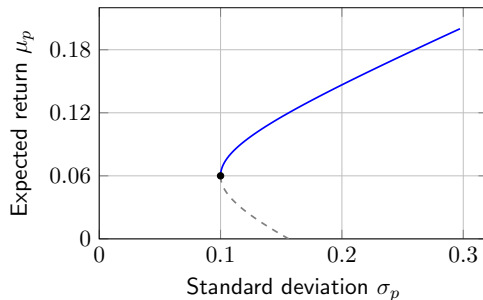
The minimum variance for expected return μ_p is:

$$\sigma_p^2(\mu_p) = \frac{1}{D}(A\mu_p^2 - 2B\mu_p + C)$$

This is a parabola in (σ_p^2, μ_p) space and a hyperbola in (σ_p, μ_p) space:

$$\frac{\sigma_p^2}{1/A} - \frac{(\mu_p - B/A)^2}{D/A^2} = 1$$

Frontier in (σ_p, μ_p) space



Illustrative constants $A = 100$, $B = 6$, $C = 0.61$ ($D = 25$). Solid: $\mu_p = \frac{B}{A} + \sqrt{\frac{D}{A} \left(\sigma_p^2 - \frac{1}{A} \right)}$ for $\sigma_p \geq 1/\sqrt{A}$; dashed: inefficient branch.

Two-Fund Separation Theorem

Theorem (Two-Fund Separation)

Every portfolio on the minimum-variance frontier can be expressed as an affine combination of any two distinct frontier portfolios.

Proof.

Frontier weights are affine in the target return: $\mathbf{w}^*(\mu_p) = \mathbf{g} + \mathbf{h}\mu_p$. Let $\mathbf{w}_1$ and $\mathbf{w}_2$ be frontier portfolios with expected returns $\mu_1 \neq \mu_2$, and set $\alpha = \frac{\mu_p - \mu_2}{\mu_1 - \mu_2}$. Then

$$\alpha \mathbf{w}_1 + (1 - \alpha) \mathbf{w}_2 = \mathbf{g} + \mathbf{h} [\alpha \mu_1 + (1 - \alpha) \mu_2] = \mathbf{g} + \mathbf{h} \mu_p = \mathbf{w}^*(\mu_p).$$

The combination is efficient only when $\mu_p \geq B/A$, the mean of the global minimum-variance portfolio. □

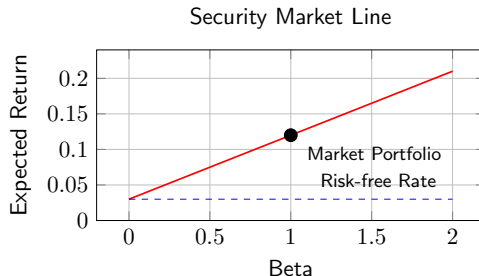
CAPM and the Security Market Line

Theorem (Capital Asset Pricing Model)

In equilibrium, the expected return of any asset i satisfies:

$$\mathbb{E}[R_i] = r_f + \beta_i(\mathbb{E}[R_M] - r_f)$$

where $\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}$.



Factor Models

Definition (K-Factor Model)

$$R_i = \alpha_i + \sum_{k=1}^K \beta_{ik} F_k + \epsilon_i$$

where R_i is the return on asset i , F_k the return on factor k , β_{ik} the loading of asset i on factor k , and ϵ_i the idiosyncratic error.

In matrix form: $\mathbf{R} = \boldsymbol{\alpha} + \mathbf{B}\mathbf{F} + \boldsymbol{\epsilon}$

Theorem (Factor Model Covariance)

$$\boldsymbol{\Sigma} = \mathbf{B}\boldsymbol{\Sigma}_f\mathbf{B}^T + \boldsymbol{\Psi}$$

where $\boldsymbol{\Psi}$ is diagonal (idiosyncratic risk).

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization**
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Von Neumann-Morgenstern Axioms

Axioms of Expected Utility

- ① **Completeness:** For any lotteries L_1, L_2 : either $L_1 \succeq L_2$, $L_2 \succeq L_1$, or both
- ② **Transitivity:** If $L_1 \succeq L_2$ and $L_2 \succeq L_3$, then $L_1 \succeq L_3$
- ③ **Continuity:** If $L_1 \succ L_2 \succ L_3$, then $\exists \alpha \in (0, 1)$ such that $L_2 \sim \alpha L_1 + (1 - \alpha)L_3$
- ④ **Independence:** If $L_1 \succeq L_2$, then $\alpha L_1 + (1 - \alpha)L_3 \succeq \alpha L_2 + (1 - \alpha)L_3$ for all L_3 and $\alpha \in (0, 1]$

Theorem (VNM Representation)

The axioms imply existence of a utility function u such that:

$$L_1 \succeq L_2 \iff \mathbb{E}[u(L_1)] \geq \mathbb{E}[u(L_2)]$$

Risk Aversion Measures

Definition (Absolute Risk Aversion)

$$A(x) = -\frac{u''(x)}{u'(x)}$$

Definition (Relative Risk Aversion)

$$R(x) = -\frac{xu''(x)}{u'(x)} = xA(x)$$

Common Utility Functions

- **CARA:** $u(x) = -\frac{1}{a}e^{-ax}$, $A(x) = a$
- **CRRA:** $u(x) = \frac{x^{1-\gamma}}{1-\gamma}$, $R(x) = \gamma$
- **HARA:** $u(x) = \frac{1-\eta}{\eta} \left(\frac{ax}{1-\eta} + b \right)^\eta$; for $b = 0$, $R(x) = 1 - \eta$

Optimal Portfolio Choice: CARA Utility

Problem Setup

Maximize: $\mathbb{E}[u(W_T)]$ where $u(w) = -\frac{1}{a}e^{-aw}$

Terminal wealth: $W_T = W_0(1 + r_f) + \mathbf{w}^T(\mathbf{R} - r_f\mathbf{1})W_0$

Theorem (CARA Optimal Portfolio)

Under multivariate normal returns $\mathbf{R} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$:

$$\mathbf{w}^* = \frac{1}{aW_0} \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f\mathbf{1})$$

Note: The dollar amounts $W_0\mathbf{w}^* = \frac{1}{a}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f\mathbf{1})$ are independent of wealth; the weights fall as $1/W_0$.

Stochastic Dominance

Definition (First-Order Stochastic Dominance)

F dominates G (written $F \succeq_{FSD} G$) if:

$$F(x) \leq G(x) \quad \forall x, \text{ with strict inequality for some } x.$$

Definition (Second-Order Stochastic Dominance)

F dominates G (written $F \succeq_{SSD} G$) if:

$$\int_{-\infty}^x [G(t) - F(t)] dt \geq 0 \quad \forall x$$

Theorem (SSD and Risk Aversion)

$F \succeq_{SSD} G$ if and only if $\mathbb{E}_F[u] \geq \mathbb{E}_G[u]$ for all nondecreasing concave u .

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory**
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Merton's Continuous-Time Model

Setup

- Asset prices: $dS_i(t) = S_i(t)[\mu_i dt + \sigma_i^T d\mathbf{Z}(t)]$, with σ_i^T the i -th row of the volatility matrix σ
- Risk-free asset: $dB(t) = rB(t)dt$
- Wealth dynamics:

$$dW(t) = [rW(t) + \pi^T(\boldsymbol{\mu} - r\mathbf{1})W(t) - c(t)]dt + W(t)\pi^T\sigma d\mathbf{Z}(t)$$

where $W(t)$ is wealth, π is the vector of portfolio weights, $c(t)$ is consumption, $\mathbf{Z}(t)$ is a vector of independent standard Brownian motions (written $\mathbf{Z}$ because W denotes wealth in this section) and $\Sigma = \sigma\sigma^T$.

Hamilton-Jacobi-Bellman Equation

Definition (Value Function)

$$J(W, t) = \max_{\{\pi, c\}} \mathbb{E} \left[\int_t^T u(c(s), s) ds + B(W(T), T) \middle| W(t) = W \right]$$

Theorem (HJB Equation)

The value function satisfies:

$$0 = \max_{\pi, c} \left\{ u(c, t) + J_t + J_W [rW + \pi^T (\mu - r\mathbf{1})W - c] + \frac{1}{2} J_{WW} W^2 \pi^T \Sigma \pi \right\} \quad (3)$$

with boundary condition $J(W, T) = B(W, T)$.

First-Order Conditions

From the HJB equation:

$$\frac{\partial u}{\partial c} = J_W \quad (\text{consumption FOC}) \quad (4)$$

$$J_W(\boldsymbol{\mu} - r\mathbf{1})W + J_{WW}W^2\boldsymbol{\Sigma}\boldsymbol{\pi} = \mathbf{0} \quad (\text{portfolio FOC}) \quad (5)$$

The optimal portfolio is:

$$\boldsymbol{\pi}^* = -\frac{J_W}{J_{WW}W}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r\mathbf{1})$$

CRRA Case: Explicit Solution

For CRRA utility $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, conjecture:

$$J(W, t) = \frac{W^{1-\gamma}}{1-\gamma} f(t)$$

Theorem (Merton's Solution)

The optimal consumption and portfolio rules are:

$$c^*(t) = W(t)f(t)^{-1/\gamma} \quad (6)$$

$$\pi^*(t) = \frac{1}{\gamma} \Sigma^{-1} (\mu - r\mathbf{1}) \quad (7)$$

where $f(t)$ satisfies an ODE.

Note: The portfolio rule is independent of wealth and time (myopic property).

Finite vs. Infinite Horizon

Infinite Horizon ($T \rightarrow \infty$)

- Constant consumption-wealth ratio c^*/W
- Constant portfolio weights

Finite Horizon

- Time-varying consumption rate
- Same portfolio weights (constant μ , r , σ)

With discount rate ρ (utility $e^{-\rho t}u(c)$), the infinite-horizon consumption-wealth ratio is

$$\frac{c^*}{W} = \frac{1}{\gamma} \left[\rho - (1 - \gamma) \left(r + \frac{(\mu - r\mathbf{1})^T \Sigma^{-1} (\mu - r\mathbf{1})}{2\gamma} \right) \right]$$

Remarks

Hedging demands arise from a stochastic investment opportunity set, not from a finite horizon. Portfolio turnpike: as the horizon grows, the optimal portfolio of an investor whose utility behaves like CRRA at large wealth converges to the CRRA portfolio.

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions**
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Fundamental Theorems of Asset Pricing

Definition (Arbitrage)

An arbitrage opportunity is a self-financing strategy with:

- Initial cost: $V_0 = 0$
- Final payoff: $V_T \geq 0$ a.s., $\mathbb{P}(V_T > 0) > 0$

Theorem (First FTAP)

In finite discrete time, no arbitrage exists if and only if there exists an equivalent martingale measure $\mathbb{Q} \sim \mathbb{P}$ such that discounted price processes are $\mathbb{Q}$ -martingales. In continuous time, the condition is no free lunch with vanishing risk (NFLVR) and $\mathbb{Q}$ is an equivalent local martingale measure.

Theorem (Second FTAP)

Assume an equivalent martingale measure exists. The market is complete if and only if that measure is unique.

Transaction Costs: No-Trade Region

Bid-Ask Spread Model

- Bid price: $S_t^b = S_t(1 - \lambda)$
- Ask price: $S_t^a = S_t(1 + \lambda)$
- Transaction cost parameter: $\lambda > 0$

Theorem (No-Trade Region)

With proportional transaction costs, the optimal strategy exhibits a no-trade region around the frictionless optimum.

To first order in λ , the boundaries satisfy:

$$\frac{\partial V}{\partial \pi_i} = \pm \lambda W \frac{\partial V}{\partial W}$$

No-Trade Region Illustration

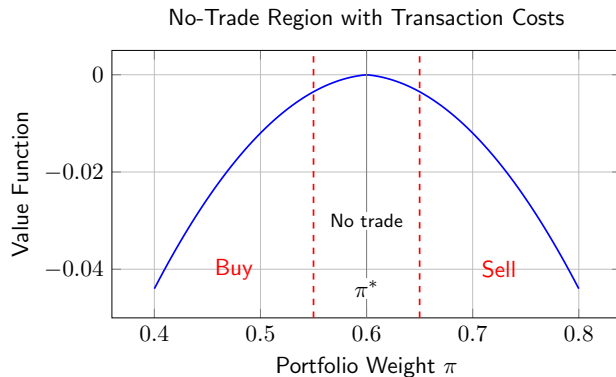


Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques**
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Convex Optimization Foundations

Definition (Convex Function)

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$.

Theorem (Jensen's Inequality)

If f is convex and X is a random variable: $f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$

KKT Conditions

Theorem (Karush-Kuhn-Tucker Conditions)

If x^ is optimal and constraint qualification holds, then $\exists \lambda^*, \nu^*$ such that:*

$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0 \quad (8)$$

$$f_i(x^*) \leq 0, \quad i = 1, \dots, m \quad (9)$$

$$h_j(x^*) = 0, \quad j = 1, \dots, p \quad (10)$$

$$\lambda_i^* \geq 0, \quad i = 1, \dots, m \quad (11)$$

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m \quad (12)$$

Robust Portfolio Optimization

Definition (Uncertainty Set)

Let $\mathcal{U}$ be a convex, compact set containing possible values of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Definition (Robust Optimization Problem)

$\min_{\mathbf{w}} \max_{(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \in \mathcal{U}} \frac{1}{2} \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}$ subject to: $\min_{(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \in \mathcal{U}} \boldsymbol{\mu}^T \mathbf{w} \geq \mu_{\min}, \mathbf{1}^T \mathbf{w} = 1$

Ellipsoidal Uncertainty Set

$$\mathcal{U} = \left\{ (\boldsymbol{\mu}, \boldsymbol{\Sigma}) : \|\boldsymbol{\mu} - \hat{\boldsymbol{\mu}}\|_{\mathbf{Q}^{-1}} \leq \kappa, \boldsymbol{\Sigma} = \hat{\boldsymbol{\Sigma}} \right\}$$

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory**
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Coherent Risk Measures

Definition (Coherent Risk Measure)

ρ is coherent if it satisfies:

- ❶ **Monotonicity:** $X \leq Y \Rightarrow \rho(X) \geq \rho(Y)$
- ❷ **Translation Invariance:** $\rho(X + c) = \rho(X) - c$
- ❸ **Positive Homogeneity:** $\rho(\lambda X) = \lambda \rho(X)$ for $\lambda \geq 0$
- ❹ **Subadditivity:** $\rho(X + Y) \leq \rho(X) + \rho(Y)$

Definition (Value-at-Risk)

$$\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) \geq \alpha\} = -q_\alpha(X)$$

Problem: VaR fails subadditivity, making it non-coherent.

Conditional Value-at-Risk

Definition (CVaR/Expected Shortfall)

$$\text{CVaR}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$$

When X has a continuous distribution, equivalently: $\text{CVaR}_\alpha(X) = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$

Theorem (CVaR Coherence)

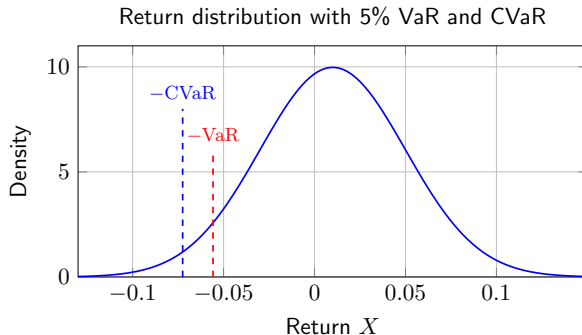
CVaR is a coherent risk measure.

Theorem (CVaR: Minimization Formula and Dual Representation)

$$\text{CVaR}_\alpha(X) = \min_{t \in \mathbb{R}} \left\{ t + \frac{1}{\alpha} \mathbb{E}[(-X - t)^+] \right\} = \sup_{Q: dQ/dP \leq 1/\alpha} \mathbb{E}_Q[-X]$$

The minimization formula makes CVaR optimization tractable via linear programming.

VaR vs CVaR Illustration



$X \sim \mathcal{N}(1\%, (4\%)^2)$, $\alpha = 5\%$: $\text{VaR}_\alpha = 5.58\%$ and $\text{CVaR}_\alpha = 7.25\%$. The dashed lines mark $-\text{VaR}_\alpha$ and $-\text{CVaR}_\alpha$ on the return axis.

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory**
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Curse of Dimensionality

The Problem

With n assets and T observations:

- Sample covariance matrix: $\hat{\Sigma} = \frac{1}{T-1} \sum_{t=1}^T (\mathbf{r}_t - \bar{\mathbf{r}})(\mathbf{r}_t - \bar{\mathbf{r}})^T$
- When $n/T \rightarrow c > 0$: $\hat{\Sigma}$ is ill-conditioned
- Traditional Markowitz fails catastrophically

Theorem (Marchenko-Pastur Law)

Suppose the entries of the return matrix are i.i.d. with true covariance $\sigma^2 \mathbf{I}$. As $n, T \rightarrow \infty$ with $n/T \rightarrow c \in (0, 1)$, the empirical spectral distribution of $\hat{\Sigma}$ converges weakly a.s. to the law with density $\rho_{MP}(\lambda) = \frac{1}{2\pi c \sigma^2} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda}$ on $[\lambda_-, \lambda_+]$, where $\lambda_{\pm} = \sigma^2(1 \pm \sqrt{c})^2$.

Shrinkage Estimators

Definition (Ledoit-Wolf Shrinkage)

$\hat{\Sigma}_{LW} = \alpha \mathbf{F} + (1 - \alpha) \hat{\Sigma}$ where $\mathbf{F}$ is a structured target and α^* minimizes expected loss.

Theorem (Optimal Shrinkage Intensity)

The optimal shrinkage intensity is: $\alpha^ = \frac{\sum_{i,j} [\text{Var}(\hat{\Sigma}_{ij}) - \text{Cov}(F_{ij}, \hat{\Sigma}_{ij})]}{\sum_{i,j} \mathbb{E}[(F_{ij} - \hat{\Sigma}_{ij})^2]}$*

Theorem (Stein's Paradox)

For $X \sim \mathcal{N}(\theta, \sigma^2 I_p)$ with $p \geq 3$, the James-Stein estimator dominates the sample mean under squared error loss.

Factor Models and PCA

Definition (Strict Factor Model)

$$\mathbf{r}_t = \boldsymbol{\alpha} + \mathbf{B}\mathbf{f}_t + \boldsymbol{\epsilon}_t$$

As stated earlier (Factor Model Covariance theorem): $\boldsymbol{\Sigma} = \mathbf{B}\boldsymbol{\Sigma}_f\mathbf{B}^T + \boldsymbol{\Psi}$ with $\boldsymbol{\Psi}$ diagonal.

Theorem (Principal Components Consistency)

Under regularity conditions, the first K principal components consistently estimate the factor space as $n, T \rightarrow \infty$.

Regularization Methods

Definition (LASSO Objective)

$$\min_{\boldsymbol{\beta}} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1$$

ℓ_1 -Regularized Portfolio

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \hat{\boldsymbol{\Sigma}} \mathbf{w} + \lambda \|\mathbf{w}\|_1 \text{ subject to: } \boldsymbol{\mu}^T \mathbf{w} = \mu_p, \mathbf{1}^T \mathbf{w} = 1$$

This promotes sparsity in portfolio weights.

Theorem (Large Portfolio Asymptotics)

If $\|\mathbf{w}\|_2 \rightarrow 0$ as $n \rightarrow \infty$ and idiosyncratic variances are bounded:
$$\text{Var}(\mathbf{w}^T \mathbf{r}) - \mathbf{w}^T \mathbf{B} \boldsymbol{\Sigma}_f \mathbf{B}^T \mathbf{w} = \mathbf{w}^T \boldsymbol{\Psi} \mathbf{w} \rightarrow 0$$

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models**
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Prospect Theory

Definition (Value Function)

$$v(x) = \begin{cases} x^\alpha & \text{if } x \geq 0 \\ -\lambda(-x)^\beta & \text{if } x < 0 \end{cases}$$

where $\alpha, \beta \in (0, 1)$ (diminishing sensitivity) and $\lambda > 1$ (loss aversion).

Definition (Probability Weighting)

$$w(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{1/\gamma}} \quad (w^+ \text{ for gains, } w^- \text{ for losses, each with its own exponent})$$

Definition (Cumulative Prospect Theory Preferences)

$V = \sum_i \pi_i v(x_i - r)$, with reference point r and rank-dependent decision weights: for gains, with outcomes ordered $x_1 \leq \dots \leq x_n$, $\pi_i^+ = w^+(p_i + \dots + p_n) - w^+(p_{i+1} + \dots + p_n)$ (analogously π_i^- with w^- for losses).

Ambiguity Aversion and Habit Formation

Definition (Max-Min Expected Utility)

$V = \min_{P \in \mathcal{P}} \mathbb{E}_P[u(W)]$ where $\mathcal{P}$ is a set of probability measures.

Definition (Smooth Ambiguity Model)

$V = \phi^{-1}(\mathbb{E}_\mu[\phi(\mathbb{E}_P[u(W)])])$ where ϕ captures ambiguity attitude and μ is second-order belief.

Definition (Habit Formation)

$u(C_t, X_t) = \frac{(C_t - X_t)^{1-\gamma}}{1-\gamma}$ where X_t is the habit level.

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects**
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

Monte Carlo Methods

Definition (Monte Carlo Estimator)

For $\theta = \mathbb{E}[g(X)]$: $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n g(X_i)$ where X_i are iid samples.

Theorem (Central Limit Theorem)

$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ where $\sigma^2 = \text{Var}(g(X))$.

Variance Reduction Techniques

- **Antithetic Variables:** Use pairs $(U, 1 - U)$
- **Control Variates:** $\hat{\theta}_{CV} = \hat{\theta} - c(\hat{\beta} - \beta)$
- **Importance Sampling:** Change of measure

Finite Difference Methods

HJB Equation Discretization

$$\frac{\partial J}{\partial t} + \max_{\pi} \{ \mathcal{L}^{\pi} J \} = 0$$

Definition (Upwind Scheme)

$$\frac{J_i^{n+1} - J_i^n}{\Delta t} + \max_{\pi} \left\{ \mu^+ \frac{J_{i+1}^n - J_i^n}{\Delta W} + \mu^- \frac{J_i^n - J_{i-1}^n}{\Delta W} + \frac{\sigma^2(\pi)}{2} \frac{J_{i+1}^n - 2J_i^n + J_{i-1}^n}{(\Delta W)^2} \right\} = 0, \text{ with}$$

$\mu^+ = \max(\mu(\pi), 0)$ and $\mu^- = \min(\mu(\pi), 0)$.

Algorithm 1 Value Iteration

Initialize V^0

repeat

$$V^{k+1}(x) = \max_{\pi} \{ u(x, \pi) + \gamma \mathbb{E}[V^k(f(x, \pi, \xi))] \}$$

until $\|V^{k+1} - V^k\| < \epsilon$

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management**
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis

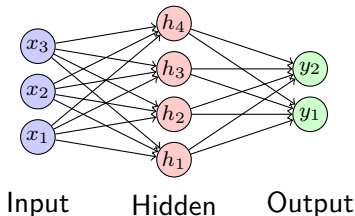
Deep Learning for Asset Pricing

Definition (Neural Network)

$f(\mathbf{x}) = W^{(L)}\sigma(W^{(L-1)}\sigma(\dots\sigma(W^{(1)}\mathbf{x} + b^{(1)})\dots) + b^{(L-1)}) + b^{(L)}$ where σ is an activation function (e.g., ReLU, tanh).

Theorem (Universal Approximation)

Neural networks with one hidden layer, sufficiently many hidden units and a non-polynomial activation can approximate any continuous function on compact sets arbitrarily well.



Reinforcement Learning

Definition (Markov Decision Process)

A tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition probabilities $P(s'|s, a)$, reward function $R(s, a)$ and discount factor γ .

Definition (Q-Function)

$$Q^\pi(s, a) = \mathbb{E}^\pi \left[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t) \middle| S_0 = s, A_0 = a \right]$$

Theorem (Bellman Equation)

$$Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s'|s, a) \max_{a'} Q^*(s', a')$$

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research**
- 14 Conclusion and Synthesis

ESG Integration

Definition (ESG-Constrained Optimization)

$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w}$ subject to:

$$\boldsymbol{\mu}^T \mathbf{w} = \mu_p \quad (13)$$

$$\mathbf{s}^T \mathbf{w} \geq s_{\min} \quad (\text{ESG constraint}) \quad (14)$$

$$\mathbf{1}^T \mathbf{w} = 1 \quad (15)$$

where $\mathbf{s}$ is the vector of ESG scores.

Shadow Price of the ESG Constraint

By the envelope theorem, the minimized objective satisfies $\frac{\partial}{\partial s_{\min}} \left(\frac{1}{2} \sigma_p^{2*} \right) = \lambda_{ESG} \geq 0$, where λ_{ESG} is the Lagrange multiplier on the ESG constraint (zero when the constraint is slack).

Cryptocurrency Portfolio Theory

Definition (Crypto Return Model)

$r_{i,t} = \alpha_i + \sum_{j=1}^k \beta_{ij} F_{j,t} + \gamma_i M_t + \epsilon_{i,t}$ where:

- $F_{j,t}$: traditional risk factors
- M_t : crypto market factor
- γ_i : crypto beta

Unique Considerations

- Extreme volatility and fat tails
- 24/7 trading (no market close, no close-to-open gap)
- Limited fundamental anchors
- Regulatory uncertainty
- Network effects and adoption dynamics

Future Research Directions

Emerging Areas

- **Quantum Computing:** Portfolio optimization with quantum algorithms
- **Climate Risk:** Integration of physical and transition risks
- **Alternative Data:** Satellite imagery, social media sentiment
- **Federated Learning:** Privacy-preserving collaborative modeling
- **Explainable AI:** Interpretable machine learning for finance

Mathematical Challenges

- Non-convex optimization in deep learning
- High-dimensional statistical inference
- Robustness to distributional shifts
- Real-time adaptive algorithms
- Integration of discrete and continuous methods

Table of Contents

- 1 Introduction and Course Overview
- 2 Mathematical Foundations
- 3 Classical Mean-Variance Theory
- 4 Utility Theory and Expected Utility Maximization
- 5 Dynamic Portfolio Theory
- 6 Incomplete Markets and Market Frictions
- 7 Advanced Optimization Techniques
- 8 Risk Measures and Coherent Risk Theory
- 9 High-Dimensional Portfolio Theory
- 10 Behavioral Finance Mathematical Models
- 11 Numerical Methods and Computational Aspects
- 12 Machine Learning in Portfolio Management
- 13 Advanced Topics and Current Research
- 14 Conclusion and Synthesis**

Key Mathematical Themes

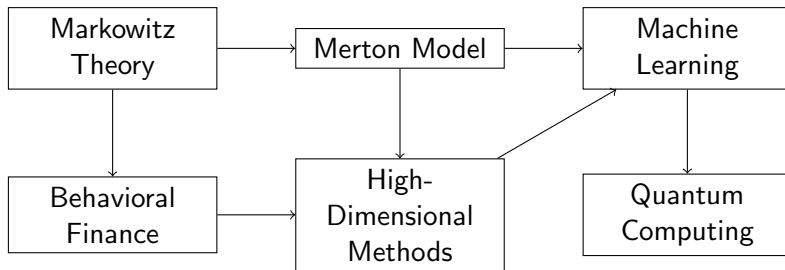
Foundational Mathematics

- **Measure Theory:** Rigorous probability foundations
- **Functional Analysis:** Optimization in infinite dimensions
- **Stochastic Calculus:** Continuous-time modeling
- **Convex Analysis:** Optimization theory and duality

Applied Methods

- **Numerical Analysis:** Computational implementation
- **Statistics:** Parameter estimation and testing
- **Machine Learning:** Pattern recognition and prediction
- **Game Theory:** Multi-agent interactions

Integration of Classical and Modern Methods



Practical Implementation Guidelines

Model Selection Hierarchy

- 1 Start with mean-variance framework
- 2 Add realistic constraints and transaction costs
- 3 Incorporate robust optimization techniques
- 4 Consider behavioral factors for retail investors
- 5 Apply machine learning for alpha generation
- 6 Use advanced risk measures for risk management

Key Takeaways

- Rigorous foundations enable robust applications
- Interdisciplinary approach combining finance, economics, psychology
- Leverage modern computing for previously intractable problems
- Always validate theoretical insights with real data

Final Remarks

Portfolio Theory Evolution

From Markowitz's mean-variance foundation to modern machine learning applications, portfolio theory continues to evolve with:

- More sophisticated mathematical tools
- Increased computational power
- Richer datasets and alternative data sources
- Growing awareness of behavioral and systemic factors

Thank you for your attention!

Questions and Discussion